

K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

November 2018

Max. Marks: 100

End Semester Exam

Duration: 3 Hrs

Class: S.Y.BTech

Name of the Course: Applied Mathematics-III

Semester: III

Course Code: UCEC301 / UITC301

Branch: COMP/IT

Instructions:

(1) All Questions are Compulsory

(2) Figures to right indicate full marks. Each sub-question has equal marks.

Question No.		Max. Marks
Q.1	A Attempt the following	
	Using Laplace transform solve $\frac{d^2y}{dx^2} + y = t$, $y(0) = 1$, $y'(0) = 0$	07
	B Attempt any THREE of the following	18
	(a) Find $L(\operatorname{erf}\sqrt{t})$	
	(b) Find the Laplace transform of $e^{-4t} \int_0^t u \sin 3u \, du$.	
	(c) Using Convolution theorem, find the inverse Laplace transform of $\frac{s}{s^4+8s^2+16}$	
	(d) Find inverse Laplace transform of $\frac{3s+1}{(s+1)(s^2+2)}$	
Q.2	(e) Express the following function as Heaviside's unit step function and find its Laplace transform $f(t) = \begin{cases} \sin t, & 0 < t < \pi \\ \cos t, & t > \pi \end{cases}$	
	A Attempt the following	06
	Obtain the Fourier series expansion of $f(x) = \left(\frac{\pi-x}{2}\right)^2$ in the interval $0 \leq x \leq 2\pi$, and $f(x+2\pi) = f(x)$.	
	B Attempt any THREE of the following	18
	(a) Find the Fourier series for $f(x) = \sin x $ in $(-\pi, \pi)$.	
	(b) Obtain the expansion of $f(x) = x(\pi-x)$, $0 < x < \pi$ as a half-range cosine series. Hence show that, $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.	
	(c) Find the complex form of Fourier series of $f(x) = \cosh 3x + \sinh 3x$, in $(-\pi, \pi)$.	
Q.3	(d) Find the Fourier transform of $f(x) = e^{- x }$.	
	A Attempt the following	08
	Find the eigen values and eigen vectors of the matrix	
	$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$	

(P.T.O.)

	B	Attempt any THREE of the following	18
	(a)	Find the eigen values of (i) $A^3 - 3A^2 + A$ (ii) $6A^{-1} + 2I$, Where $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$	
	(b)	Verify Cayley Hamilton theorem and find A^{-1} , where $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 4 \\ 3 & 1 & -1 \end{bmatrix}$	
	(c)	If $\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$, prove that $A^{50} - A^{49} = \begin{bmatrix} -4 & 4 \\ 2 & -2 \end{bmatrix}$.	
	(d)	Is the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ diagonalisable? If so find the diagonal form and the transforming matrix.	
	(e)	Determine whether matrix $A = \begin{bmatrix} 2 & -3 & 3 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix}$ is derogatory and if so, find its minimal polynomial.	
Q.4	A	Attempt the following	07
		If $\vec{f} = x^2\vec{i} + xz\vec{j} + yz\vec{k}$ and $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$, find $\text{div}(\vec{f} \times \vec{r})$ and $\text{curl}(\vec{f} \times \vec{r})$.	
	B	Attempt any THREE of the following	18
	(a)	Find the directional derivative of $\phi = xy^2 + yz^2$ at $(2, -1, 1)$ in the direction normal to the surface $x^2y + y^2x + yz^2$ at $(1, 1, 1)$.	
	(b)	Prove that $\nabla f(r) = f'(r) \frac{\vec{r}}{r}$, where $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$. Hence find f if $\nabla f(r) = -\frac{\vec{r}}{r^5}$ and $f(1) = 0$.	
	(c)	Show that the vector $\vec{f} = (x^2 - yz)\vec{i} + (y^2 - xz)\vec{j} + (z^2 - yx)\vec{k}$ is irrotational and hence find its scalar potential.	
	(d)	Evaluate by Green's theorem $\oint (3x^2 - 8y^2)dx + (4y - 6xy)dy$, along C, where C is the boundary of the region bounded by $x = y^2$ and $y = x$.	
	(e)	Use Gauss's Divergence theorem to evaluate $\iint \vec{N} \cdot \vec{F} ds$, where $\vec{F} = 2x\vec{i} + xy\vec{j} + z\vec{k}$, over the region bounded by the cylinder $x^2 + y^2 = 4, z = 0, z = 6$.	

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End Semester Exam
Nov - Dec 2018

Max. Marks: 100

Class: SY B.Tech

Name of the Course: Object Oriented Programming Methodology

Course Code: UCEC 302

Duration: 3 hrs

Semester: III

Branch: Computer

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Explain System.arraycopy() with example	05
Q 1 (b)	Explain the architecture of JVM with the help of diagram.	05
Q2 (a)	Create a class of Course with course code, name, and number of students as a data fields. Class Department should have name and list of courses offered. Write a program to maintain a list of courses offered by department using vector. And perform following operations in class Department 1. Add a course 2. Delete course 3. Display courses (Make use of enumeration for displaying the elements of vector.)	10
Q2 (b)	Write a program to implement the Rock, Paper, Scissor game. Numbers of Players are two, In which one is computer. For Computer player make use of Random function. And for other user accepts Rock /paper/Scissor as input from user. Number of turns are 10. After completing the 10 turns, Display the winner of the game. At the beginning of the game accept name of user and at every turn update the score of each user.as soon as you create the instances of user it should initialize the score to 0 by using constructor.	10
Q.2 (c)	Write a program to find different 'permutation and combinations of the given string. (Make use of String functions)	10
	OR	
	Write a program to count the number of words, capital letters and small case letters in the given string. (Make use of String functions)	10
Q3 (a)	Draw the class diagram for Hospital management system. Assume suitable data. Show the relations between different classes and label the relations properly. Also show the multiplicity according to your assumption. (Specify the assumptions clearly)	10

Q3 (b)	How the aggregation and Composite relationship between classes are implemented explain with example.	10
Q4 (a)	Explain the ArrayList in java along with its any 5 functions. Give an example for creating an ArrayList and make use of Iterator for displaying the elements of ArrayList.	10
Q4 (b)	What is Error and Exception in Exception hierarchy? Explain the use of throws in exception handling with example. OR Write a program to read and display the content of text file. Program should handle exceptions which may occur while performing this operation. Your program should close the file at the end irrespective of exception occur or not.	10 (4+6) 10
Q5 (a)	Write short note on (any two) 1. wrapper class 2. Packages 3. life cycle of threads 4. Static keyword	10(5+5)
Q5 (b)	Explain life cycle of an applet. Write a program to display house using applets. OR Write a program to display smiley face filled with color. After every 5 sec, program should change the color of face.	10 10

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End Semester Exam

Nov – Dec 2018

Max. Marks: 100

Class: SY B Tech

Name of the Course: Data Structures

Course Code: UCEC303

Duration: 3 hrs

Semester: III

Branch: CMPN

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary
- (4) Write program / algorithm wherever applicable.

Question No.		Max. Marks
Q 1	Answer any five questions	25
Q 1 (a)	Discuss in brief the advantages of using linked lists over arrays.	
Q 1 (b)	What is use of ADT? Give an example for the Stack ADT.	
Q 1 (c)	Explain breadth first search with an example.	
Q 1 (d)	Give an example of expression trees as an application of the tree data structure. How is it useful?	
Q 1 (e)	What are the applications of searching and sorting?	
Q 1 (f)	What is recursion? How can you use recursion to compute sum of n natural numbers? Give function for the same.	
Q 1 (g)	Explain priority queue as an application of the heap data structure	
Q2 (a)	Write a program to implement a Doubly Linked List which performs the following operations: (i) Inserting element in the beginning (ii) Inserting element in the end (iii) Deleting a particular element (iv) Displaying the list OR Explain with a program how simulation can be carried out using a Queue Data Structure.	10 10
Q 2 (b)	Write a program for DFS traversal of a graph. Explain its working with an example.	10
Q 2 (c)	What is a double ended queue? What are its variants? Explain in brief.	05
Q3 (a)	Write a program to evaluate a prefix expression. Explain the working with an example. OR Write a program to implement a Circular queue. The program should be able to perform the following operations: (i) Creating the queue (ii) Deleting from the queue (iii) Inserting in the queue (iv) Displaying the current elements of the queue	10 10

Q 3 (c)	How can stack be represented as linked list?	05
Q 4 (a)	What is Hashing? Insert the following values in a hash table of size 10 using Quadratic Probing. Also find the number of collisions. 23, 13, 27, 65, 56, 41, 19, 26, 83, 67 OR Write a menu driven program to implement Insertion Sort and Selection sort. The program should ask the user to enter n elements and display the option for either insertion sort or selection sort. Depending upon the selection, the program should sort the elements.	10
Q 4 (b)	What is an AVL tree? Explain with example. Also explain the different rotations that are used while inserting an element into the tree.	10
Q 4 (c)	How can you solve sparse matrix addition using linked lists?	05
Q.3(b)	Create a BST for following numbers 26, 38, 15, 29, 33, 12, 20, 31, 18, 30 Write preorder and postorder traversal sequence of the same.	10

27-11-2018 (E)

K. J. Somaiya College of Engineering, Mumbai-77
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End Semester Exam
Nov – Dec 2018

Duration: 3 Hrs
Semester: III
Branch: COMP

Max. Marks: 100

Class: S.Y.B.Tech

Name of the Course: Computer Organization and Architecture

Course Code: UCEC304

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Define computer organization and computer architecture. Draw a neat diagram with explanation of the expanded structure of IAS computer developed by Von Neumann.	10
Q 1 (b)	Represent $(-0.001750)_{10}$ in single precision and in double precision format.	10
Q2 (a)	What are the advantages of Booth's recoding algorithm over Booth's algorithm? Solve $(9) \times (6)$ using Booth's recoding algorithm.	10
Q2 (b)	Explain restoring division method for signed numbers. Hence perform $(-7)/(-3)$.	10
Q3 (a)	What is cache coherence problem? How it is overcome by using MESI protocol in uniprocessor and multiprocessor system?	10
Q3 (b)	What is memory interleaving? Compare and contrast its types.	10
Q4 (a)	List and explain registers of 8086.	10
Q4 (b)	Write the microprogram for $ADD R1, (R3)$. Explain significance of each step along with the role of register involved.	10
Q4(a)	Draw instruction cycle state diagram. Explain execution of instruction in detail.	10
Q4(b)	Explain different methods to design control unit of computers. Differentiate between them.	10
Q5 (a)	Explain programmed I/O. What are two modes of addressing of programmed I/O?	10
Q5 (b)	What are pipeline hazards? How to avoid them?	10
OR		
Q5(a)	What is an Interrupt? What processor does if an interrupt occurs? Explain Interrupt driven I/O?	10
Q5(b)	What is pipelining? Discuss various design issues of pipeline architecture.	10

29.11.2018 (E)

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End Semester Exam

Nov- Dec 2018

Max. Marks: 100

Class : S . Y . B T E C H

Name of the Course: Discrete Structure and Graph Theory.

Course Code: UCEC305

Duration: 3Hours

Semester: III

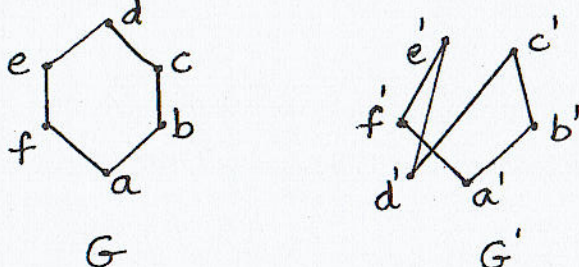
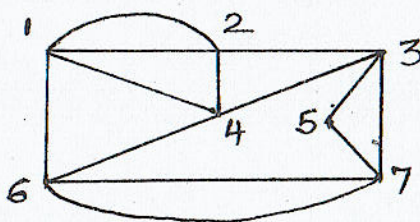
Branch: COMP

Instructions:

1. All Questions are Compulsory

2. Assume suitable data wherever necessary.

Question No.		Max. Marks
Q 1	a) Show that $1^2 + 2^2 + 3^2 + \dots + n^2 = n(n+1)(2n+1)/6$, $n \geq 1$ by mathematical induction.	5
	b) If $f: \mathbb{R} \rightarrow \mathbb{R}$ is given by $y=2x+1$, Prove that f is one to one and onto.	5
	c) Find how many integers between 1 and 60 are not divisible by 2,3 and 5 respectively.	5
	d) Solve the recurrence relation $a_n = a_{n-1} + 2a_{n-2}$, $n \geq 2$ with initial conditions $a_0 = 0$, $a_1 = 1$.	5
Q 2	a) Let R be a relation on A . Prove that i) If R is reflexive, so is R^{-1} . ii) R is symmetric if and only if $R = R^{-1}$. iii) R is antisymmetric if and only if $R \cap R^{-1} \subseteq I_A$.	10
	b) If 11 numbers between 1 and 20 are chosen, show that at least two of them will be multiples of each other.	5
	c) Show that the maximum number of edges in a simple graph with n vertices is $n(n-1)/2$.	5
Q 3	a) Let $A = \{1,2,3,4\}$ and relation $R = \{(1,4), (2,1), (2,3), (3,1), (3,4) \text{ and } (4,3)\}$ State Warshall's algorithm and Find the transitive closure of R by Warshall's Algorithm.	10
	b) Test whether the following function is injective $F: \mathbb{Z} \rightarrow \mathbb{Z}$, $f(x) = x^2 + x + 1$	05

	c) Let A be a set of integers and let R be a relation on $A \times A$ defined by $(a,b) R (c,d)$ if $a+d=b+c$. Prove that R is an equivalence relation.	05
OR		
Q 3	a) If R is a relation from A to B and S is a relation from B to C. Show that $(R \circ S)^{-1} = S^{-1} \circ R^{-1}$.	5
	b) Consider the poset $A = (\{1, 2, 3, 4, 6, 9, 12, 18, 36\}, /)$ find the greatest lower bound and the least upper bound of the sets $\{6, 8\}$ and $\{4, 6, 9\}$.	5
	c) Obtain the conjunctive normal form of $(\sim p \wedge q \wedge r) \vee (p \wedge q)$	5
	d) Check whether $A = \{2, 4, 12, 16\}$ is lattice under divisibility.	5
Q 4	a) Determine whether the following graphs are isomorphic? 	5
	b) A connected planar graph has 10 vertices each of degree 3. Into how many regions, does a representation of this planar graph split the plane?	5
	c) If eight persons are chosen from any group, show that at least two of them will have the same birthday <i>in a week</i> .	5
	d) Define Eulerian path and circuit. Find Eulerian path and circuit. 	5

Q 5	a) Consider the $(2, 6)$ group encoding function $e : B^2 \rightarrow B^6$ defined by $e(00) = 000000$ $e(01) = 011110$ $e(10) = 101101$ $e(11) = 110011$ Decode the following relative to maximum likelihood decoding function. i) 001110 ii) 111101 iii) 110010	10
	b) Prove that if $a^2 = a$, then $a=e$, a being an element of a group.	05
	c) Show that the set $\{1,2,3,4,5\}$ is not a group under addition and multiplication modulo 6.	05
OR		
Q 5	a) Show that the $(3, 6)$ encoding function $e : B^3 \rightarrow B^6$ defined by $e(000) = 000000$ $e(100) = 100101$ $e(001) = 000110$ $e(101) = 100011$ $e(010) = 010010$ $e(110) = 110111$ $e(011) = 010100$ $e(111) = 110001$ is a group code.	10
	b) Let $H =$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ be a parity check matrix, Determine the group code $e_H : B^3 \rightarrow B^6$	10